

Shakespeare Text Generation with RNN

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Abstract. This report presents a comprehensive implementation and analysis of Recurrent Neural Networks (RNNs) for next-word prediction and text generation using the Shakespeare dataset. A custom LSTM model with word-level embeddings was successfully trained to achieve a validation accuracy of 20.30% and perplexity of 200.22. A systematic ablation study on model configurations was conducted, demonstrating the impact of architectural choices on text generation performance and overfitting patterns. The final model generates Shakespearean-style text from seed phrases with moderate quality. Insights and future directions are discussed.

1 Introduction

Next-word prediction using Recurrent Neural Networks (RNNs) is a central task in natural language processing with applications in text generation, language modeling, and conversational agents. This report focuses on training an RNN model on a classic literary corpus, the Shakespeare text dataset, for generating coherent texts in Shakespearean style. The implementation includes custom word embeddings trained from scratch and a thorough evaluation of different model architectures and hyperparameters.

2 Problem Statement

The objective is to implement and train an RNN model capable of predicting the next word given a preceding sequence, specifically on Shakespeare's plays and sonnets. The model should generate coherent text from a seed phrase, evaluated using accuracy and perplexity metrics. An ablation study is included to analyze the effects of model depth, hidden layer size, and dropout. The Shakespeare dataset from Hugging Face, consisting of over one million characters and 200,000 words, is used for training.

3 Technical Implementation

3.1 Development Environment

The experiments were conducted on Google Colab with access to a Tesla T4 GPU supporting CUDA acceleration. The implementation utilized PyTorch with

several supporting libraries for dataset handling and visualization, including `datasets`, `transformers`, `matplotlib`, and `seaborn`. Random seeds were fixed to ensure reproducibility.

3.2 Dataset Integration and Preprocessing

The Tiny Shakespeare corpus from Hugging Face contains approximately 1,115,394 characters and 202,651 words. Text preprocessing involved:

- Lowercasing and whitespace normalization
- Filtering to retain relevant characters
- Tokenization into word-level tokens with a vocabulary size of 3,000 covering 82.9% of the corpus
- Padding and truncation to fixed-length sequences of 20 words
- Splitting data into 80% training and 20% validation subsets

3.3 Model Architecture

The model comprises:

- **Embedding Layer:** Custom word embeddings of dimension 128 with dropout of 0.15
- **RNN Core:** Two-layer LSTM with 256 hidden units per layer, inter-layer dropout, and batch-first processing
- **Output Layer:** Dropout rate of 0.3 followed by a linear layer projecting to the vocabulary size (3,000)

Weights were initialized using Xavier uniform distribution, and biases to zero.

3.4 Training Configuration

Training was performed over 15 epochs with batch size 64 using the Adam optimizer at a learning rate of 0.001. The learning rate was scheduled using `ReduceLROnPlateau` to decrease on validation loss plateaus. Cross-entropy loss was used, and gradient clipping at max norm 1.0 prevented exploding gradients.

4 Training Results and Performance

4.1 Training Dynamics

The model displayed three training phases:

1. Initial learning (epochs 1-2) with rapid loss and perplexity reductions
2. Optimization phase (epochs 3-6) with steady improvements and best validation results at epoch 5
3. Overfitting phase (epochs 7-15) where validation loss stagnated or increased despite training loss reduction

Table 1. Validation Performance at Select Epochs

Epoch	Val Loss	Val Accuracy (%)	Val Perplexity
1	5.67	17.78	288.76
5	5.30	20.30	200.22
10	5.46	20.15	235.31
15	5.59	20.28	268.66

4.2 Text Generation

The trained model generated text continuations from seed phrases with context windows of 20 words and output sequences of 12 tokens. Sampling used temperature 0.8 and top-k filtering with 50 tokens to balance creativity and coherence. The generated text maintained Shakespearean themes, characters, and style but occasionally showed syntactic and semantic inconsistencies.

5 Ablation Study

Five model configurations tested modifications in hidden size, number of layers, and dropout rate. Key findings:

- A hidden size of 256 provided the best balance between performance and parameter efficiency.
- Increasing layers beyond two caused overfitting and degraded performance.
- Higher dropout (0.5) hindered learning compared to 0.3.

Table 2. Ablation Study Results

Configuration	Val Accuracy (%)	Val Perplexity	Parameters
Baseline (2 layers, 256 hidden, 0.3 dropout)	20.30	200.22	2,076,600
Smaller Hidden (128)	19.54	202.29	1,035,192
Larger Hidden (512)	19.97	193.66	5,339,064
Higher Dropout (0.5)	19.60	194.77	2,076,600
More Layers (3 layers)	19.01	204.21	2,602,936

6 Challenges and Solutions

- **Dataset loading:** Handled multiple formats and fallback methods to ensure reliable data access.
- **Overfitting:** Detected clear divergence between training and validation metrics; recommended early stopping around epoch 5-6.
- **Training stability:** Gradient clipping prevented exploding gradients.
- **Text coherence:** Balancing temperature and top-k filtering improved generation quality.

7 Conclusion

This project implemented a robust word-level LSTM model for Shakespeare text generation achieving 20.30% validation accuracy and 200.22 perplexity. The architecture and training strategies effectively captured stylistic and lexical patterns in the dataset. The ablation study provided valuable insights on model configuration choices. Future work includes exploring attention mechanisms, transformer architectures, and improvements in text coherence and semantic understanding.